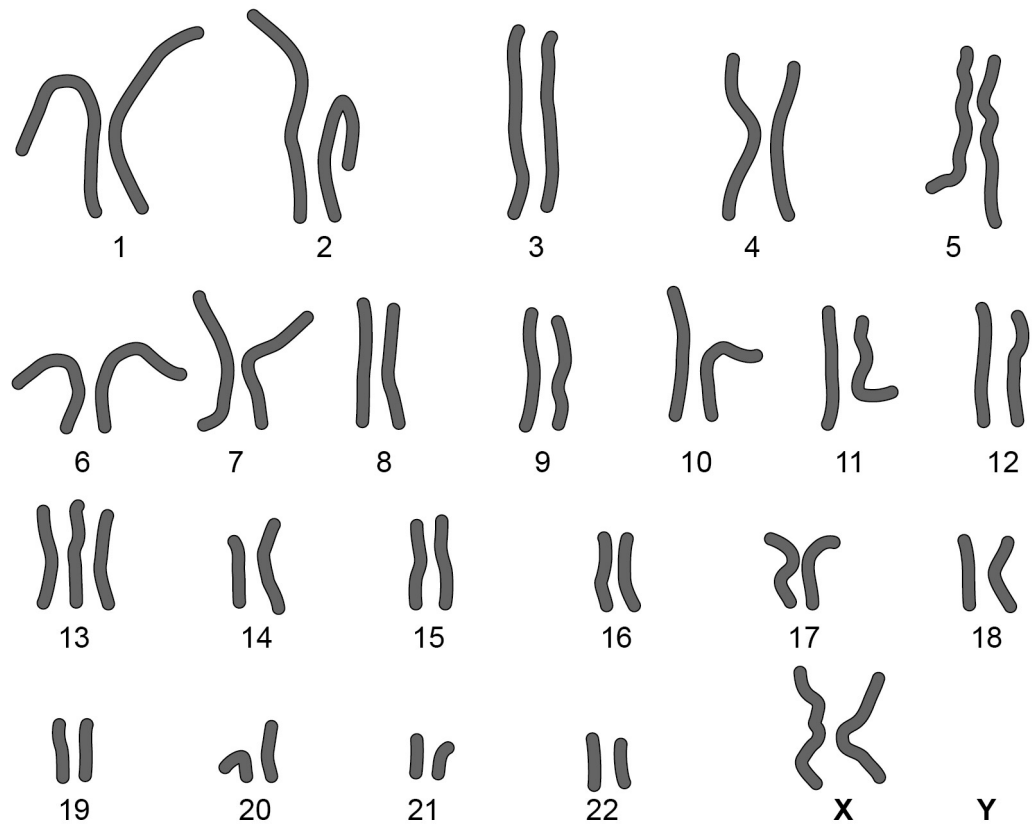


0 8

Patau syndrome is a condition caused by a mutation affecting chromosome number. All the cells of the body will have this mutation.

Figure 8 shows the chromosomes from one of the cells of a female who has Patau syndrome.

Figure 8



0 8 . 1

What is the effect of Patau syndrome on the chromosomes of this female?

[1 mark]



0 8 . 2

Describe how the change in chromosome number in Patau syndrome was produced.

[2 marks]

0 8 . 3

Explain why all the cells of the body will have this mutation.

[2 marks]

Question 8 continues on the next page

Turn over ►

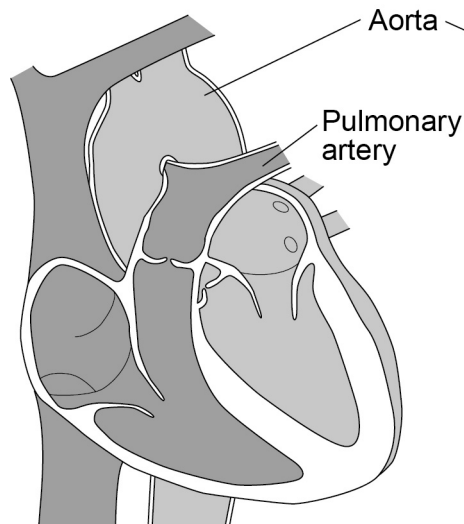
Most children born with Patau syndrome die in the first 12 months, often due to defects of circulation of blood.

One of these defects is patent ductus arteriosus (PDA). This can result in some of the blood flowing between the aorta and the pulmonary artery.

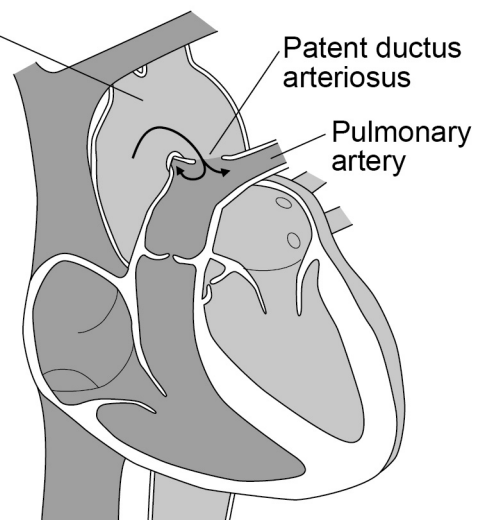
Figure 9 shows a healthy child's heart and the heart of a child with PDA.

Figure 9

Healthy child's heart



Heart of a child with PDA



0 8 . 4

Suggest how the flow of some of the blood between the aorta and pulmonary artery could cause children to die in the first 12 months.

[3 marks]

[Extra space]

